

# Jump Modelling with Dirichlet Process Mixture

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# Motivation

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# Why We Interested in Jumps?

- **Empirical stylized facts of financial returns** (Cont, 2001):  
fat-tails
- **Empirical option implied volatility (IV) smile/smirk**
- **Market turbulence:** such as 2008 GFC, 2020 pandemic

# Jump Diffusion (JD) Models

## JD models are introduced to:

- Model the sudden and extreme price changes that is not captured by volatility models,
- Capture the fat tails of return distribution,
- Better explain the empirical IV smile (Pan, 2002; Eraker *et al.*, 2003; Eraker, 2004).

## JD model structure:

1. Jump occurrence:  $\Delta N(\delta)$ ,
2. Jump size:  $Z \sim F(\cdot)$ .

Early literature on JD models: Merton (1976), Bates (1996), Kou (2002), Kou (2007).

Further investigation of jump structure: dynamic jumps (Maheu and McCurdy, 2004); jump clustering (Aït-Sahalia *et al.*, 2015); jump directions (Maneesoonthorn *et al.*, 2017).

## Parametric SVJ model:

- Assuming a particular distribution for jump size, usually normal or exponential distribution;
- Not flexible enough if considering both positive and negative jumps
- The misspecification in jump distribution will be reflected in less accurate option prices and IV.

**Our proposed DPM jump model aims to:**

- Model the jump size distribution flexibly;
- Capture both negative and positive jumps;
- Recover the empirical jump size distribution;

# Model Specification

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## Model Features:

1. Jump is assumed to be i.i.d.,
2. Jump intensity  $\delta$  is constant over time,
3. Only price jump is considered in the model.



# Full Model Specification

On day  $t$ , let  $r_t$  be the log return and  $h_t$  be the log-variance of return and  $\Delta N_t$  represents the jump arrival and  $Z_t$  is the jump size.

**Measurement equation:**

$$r_t = \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \quad (1)$$

**State equation:**

$$\begin{aligned} h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_t^h \varepsilon_t^h, \\ \Delta N_t &\sim \text{Bernoulli}(\delta), \\ Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2), \\ w_k &= v_k \prod_{l=1}^{k-1} (1 - v_l), \text{ with } v_k \sim \text{Beta}(1, \alpha), \end{aligned} \quad (2)$$

where  $\varepsilon_t^r, \varepsilon_t^{BV}, \varepsilon_t^h$  follow  $N(0,1)$ . We model jump size  $Z_t$  distribution as a stick-breaking process (SBP) (Sethuraman, 1994).

# Inference

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Let  $\Theta = \{\phi_0, \phi_1, \sigma_h^2, \sigma_{BV}^2\}$  and  $\Phi = \{\mathbf{w}, \mu, \sigma^2, \delta\}$  where  $\mu = \{\mu_1, \dots, \mu_\infty\}$  and  $\sigma^2 = \{\sigma_1^2, \dots, \sigma_\infty^2\}$ . The joint posterior can be represented as

$$\begin{aligned} p(\Theta, \Phi, \mathbf{h}, \Delta \mathbf{N}, \mathbf{Z} | \mathbf{r}) &\propto p(\phi_0, \phi_1 | \sigma_h^2) p(\sigma_h^2) p(\delta) p(h_0) \\ &\quad \prod_{t=1}^T \left[ p(\Delta N_t | \delta) p(h_t | h_{t-1}, \phi_0, \phi_1, \sigma_h^2) \right. \\ &\quad \left. p(r_t | h_t, Z_t, \Delta N_t) \prod_{k=1}^{\infty} p(\mu_k, \sigma_k^2) p(w_k) p(Z_t | \mu_k, \sigma_k^2) \right] \end{aligned} \quad (3)$$

An auxiliary variable  $u_t$  is introduced who is uniformly distributed conditional on  $\mathbf{w}$  and indicator variable of jump mixture  $S_t^Z$ .

$$u_t | S_t^Z, \mathbf{w} \sim U(0, w_{S_t^Z}). \quad (4)$$

The joint distribution of  $u_t$  and  $Z_t$  is:

$$\begin{aligned} p(u_t, Z_t | \cdot) &= \sum_{k=1}^{\infty} w_k U(u_t | 0, w_k) N(Z_t | \mu_k, \sigma_k^2), \\ &= \sum_{k=1}^{\infty} I(u_t < w_k) N(Z_t | \mu_k, \sigma_k^2), \\ &= \sum_{k \in A_t} N(Z_t | \mu_k, \sigma_k^2) \end{aligned} \quad (5)$$

where  $A_t = \{k : u_t < w_k\}$ . Notably,  $A_t$  here is a set with finite elements (Neal, 2003; Walker, 2007).

# MCMC Algorithm

With total  $M$  iterations, the MCMC algorithm is:

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## Algorithm 1 Full Gibbs sampling algorithm

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**Require:** prior,  $\mathbf{r}$

Initialize ( $i = 0$ )  $\Theta, \mathbf{S}$  for initialization of  $\mathbf{h}$ ;

Initialize ( $i = 0$ )  $(\mu_{S_z^z}, \sigma_{S_z^z}^2), \mathbf{S}^z$  for initialization of  $\mathbf{Z}$ .

Initialize ( $i = 0$ )  $\delta$

For each MCMC iteration  $i$

1. Sample  $\Delta \mathbf{N}^{(i)} | \mathbf{h}^{(i-1)}, \mathbf{r}, \mathbf{Z}^{(i-1)}, \delta^{(i-1)}$
2. Sample  $\delta^{(i)} | \Delta \mathbf{N}^{(i)}$
3. Sample  $\mathbf{w}^{(i)} | \mathbf{S}_z^{(i-1)}, \mathbf{u}^{(i)} | \mathbf{w}^{(i)}, \mathbf{S}_z^{(i-1)}$
4. Sample  $(\mu_{S_z^z}, \sigma_{S_z^z}^2)^{(i)} | \mathbf{S}_z^{(i-1)}, \mathbf{Z}^{(i-1)}$
5. Sample  $\mathbf{S}_z^{(i)} | \mathbf{Z}^{(i-1)}, (\mu_{S_z^z}, \sigma_{S_z^z}^2)^{(i)}, \mathbf{w}^{(i)}, \mathbf{u}^{(i)}$
6. Sample  $\mathbf{Z}^{(i)} | \mathbf{h}^{(i-1)}, \mathbf{r}, \Delta \mathbf{N}^{(i)}, \mathbf{S}_z^{(i)}, (\mu_{S_z^z}, \sigma_{S_z^z}^2)^{(i)}$
7. Sample  $\mathbf{h}^{(i)} | \mathbf{r}, \Delta \mathbf{N}^{(i)}, \mathbf{Z}^{(i)}, \mathbf{S}^{(i-1)}, \Theta^{(i-1)}$   $\triangleright$  using 7-component mixture (Kim *et al.*, 1998).
8. Sample  $\Theta^{(i)} | \mathbf{h}^{(i)}$
9. Sample  $\mathbf{S}^{(i)} | \mathbf{h}^{(i)}, \mathbf{r}$   $\triangleright$   $\mathbf{S}$  is the indicator for seven point mixture.
10. Go to step (1) and repeat for  $M$  iterations.

**return**  $\mathbf{h}, \mathbf{Z}, \Delta \mathbf{N}, \Theta$

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## **Simulation: Model Sensitivity**

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# Simulation Analysis

The purpose of this simulation is to investigate the model performance:

- when true jump intensity changes;
- when true jump size changes;
- how our model performs compared with parametric SVJ model where jump size  $Z_t \sim N(\mu_z, \sigma_z^2)$ .

We compare the model performance in terms of:

- Sensitivity = probability of a model-identified jump given there is a true jump;
- Specificity = probability of a model-identified no jump given there is no jump.

# True Data Generating Process

We generate  $T = 2000$  observations from the following true DGP:

$$\begin{aligned}r_t &= \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \\h_t &= -0.06 + 0.94h_{t-1} + 0.38\varepsilon_t^h, \\ \Delta N_t &\sim \text{Bernoulli}(\delta), \\ Z_t &\sim 0.6 \times TN(-\mu_z, 0.8, -\infty, -\mu_z) + 0.4 \times TN(\mu_z, 0.6, \mu_z, +\infty).\end{aligned}\tag{6}$$

We compare the model performance on a various range of  $\delta = \{5\%, 10\%\}$  and  $\mu_z = \{2\text{-}\sigma, 2.5\text{-}\sigma\}$  away from the conditional mean of return distribution.



# Jump Size Sensitivity Analysis I

| Jump Size Sensitivity Analysis |                                   |                                   |
|--------------------------------|-----------------------------------|-----------------------------------|
|                                | 2- $\sigma$                       | 2.5- $\sigma$                     |
| Sensitivity                    |                                   |                                   |
| Parametric                     | <b>0.6375</b><br>(0.5882, 0.6845) | <b>0.7838</b><br>(0.7576, 0.8131) |
| DPM                            | 0.6114<br>(0.5722, 0.6524)        | 0.7587<br>(0.7323, 0.7879)        |
| Specificity                    |                                   |                                   |
| Parametric                     | 0.8855<br>(0.8635, 0.9060)        | 0.8845<br>(0.8663, 0.9012)        |
| DPM                            | <b>0.9469</b><br>(0.9313, 0.9619) | <b>0.9634</b><br>(0.9553, 0.9711) |

Table 1: Performance of parametric and DPM models as the **true jump size changes**. The true jump intensity  $\delta = 10\%$  and  $\mu_z$  changes from 2- $\sigma$  to 2.5- $\sigma$  away from the conditional mean of return. The post-convergence average and 90% posterior interval of correctly identified jump proportion (sensitivity) and the correctly identified no-jump proportion (specificity) are provided in the table.

# Jump Intensity Sensitivity Analysis I

| Jump Intensity Sensitivity Analysis |                                   |                                   |
|-------------------------------------|-----------------------------------|-----------------------------------|
|                                     | $\delta = 5\%$                    | $\delta = 10\%$                   |
| Sensitivity                         |                                   |                                   |
| Parametric                          | <b>0.8312</b><br>(0.7912, 0.8681) | <b>0.9038</b><br>(0.8827, 0.9248) |
| DPM                                 | 0.8281<br>(0.7912, 0.8571)        | 0.8968<br>(0.8761, 0.9159)        |
| Specificity                         |                                   |                                   |
| Parametric                          | 0.9578<br>(0.9497, 0.9680)        | 0.9172<br>(0.9056, 0.9293)        |
| DPM                                 | <b>0.9918</b><br>(0.9890, 0.9948) | <b>0.9850</b><br>(0.9817, 0.9884) |

Table 2: Performance of parametric and DPM models as the **true jump intensity changes**. The true jump size is fixed at  $3.5\text{-}\sigma$  away from the conditional mean of return. The true jump intensity changes from 5% to 10%. The post-convergence average and 90% posterior interval of correctly identified jump proportion (sensitivity) and the correctly identified no-jump proportion (specificity) are provided in the table.

Estimated density of detected jumps:

- Step 1: For each iteration  $i$ , we obtain detected jumps:  
 $\mathbf{Z}^{(i)} | \Delta \mathbf{N}^{(i)} == 1;$
- Step 2: We fit kernel density to detected jumps;
- Step 3: We then obtain the average density of detected jumps across all MCMC iterations.

# Jump Size Sensitivity Analysis II: Detected Jump Kernel

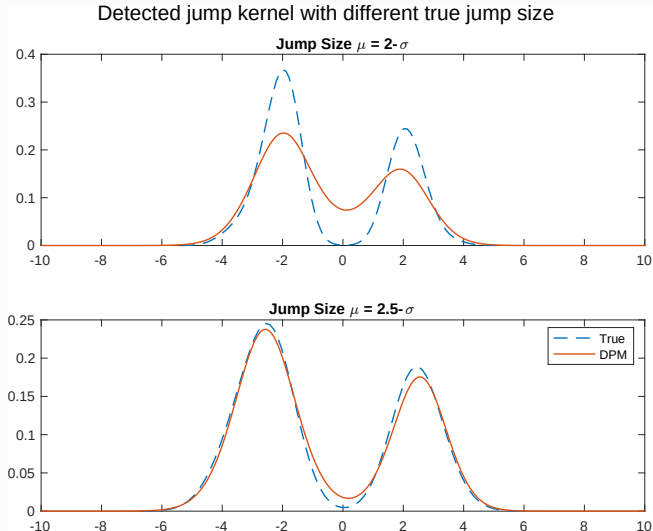


Figure 1: The estimated jump kernel for DPM jump model under different true DGPs. The blue dashed line represents the true jump density. The true jump size changes from  $2\sigma$  to  $2.5\sigma$  away from the conditional mean of the return, while the jump intensity is fixed at 10%.

# Jump Intensity Sensitivity Analysis II: Detected Jump Kernel

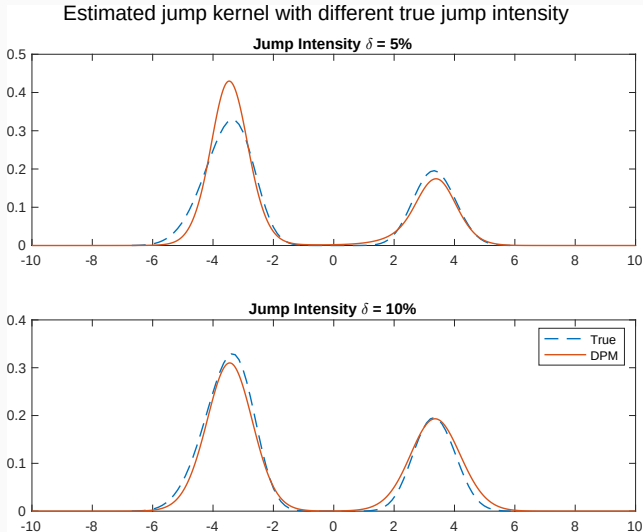


Figure 2: The estimated jump kernel for DPM jump model under different true DGPs. The blue dashed line represents the true jump density. The true jump intensity  $\delta$  changes from 5% to 10% while the true jump size is fixed at  $3.5\text{-}\sigma$  away from the conditional mean.

# Preliminary Empirical Analysis

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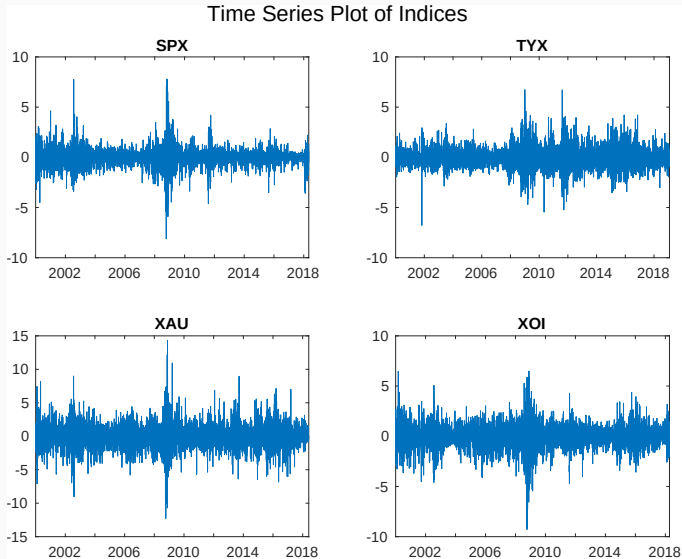


Figure 3: Time series plots of four financial indices. SPX - S&P 500 index, TYX - Treasury yield index (30 year bond), XAU - gold and silver index and XOI - oil index.

# Estimated jump size distribution

| Estimated jump size distributions for financial indices |                             |                                      |
|---------------------------------------------------------|-----------------------------|--------------------------------------|
| Parameters                                              | Posterior Mean/Mode (for K) | 90% Posterior Interval/Range (for K) |
| <b>SPX</b>                                              |                             |                                      |
| K                                                       | 1                           | [1,4]                                |
| $\mu_z$                                                 | -0.0430                     | (-0.1920, 0.1006)                    |
| $\sigma_z$                                              | 0.7037                      | (0.5747, 0.8380)                     |
| <b>TYX</b>                                              |                             |                                      |
| K                                                       | 1                           | [1,10]                               |
| $\mu_z$                                                 | 0.0683                      | (-0.0662, 0.2222)                    |
| $\sigma_z$                                              | 0.9142                      | (0.7808, 1.0749)                     |
| <b>XAU</b>                                              |                             |                                      |
| K                                                       | 2                           | [2,10]                               |
| $\mu_z$                                                 | 2.2897; -2.0587             | (1.7668, 2.7564); (-2.4320, -1.6994) |
| $\sigma_z$                                              | 0.6277; 0.6128              | (0.4329, 0.8831); (0.4053, 0.9377)   |
| <b>XOI</b>                                              |                             |                                      |
| K                                                       | 2                           | [2,10]                               |
| $\mu_z$                                                 | 0.9448; -1.6652             | (0.6030, 1.2632); (-2.0360, -1.2617) |
| $\sigma_z$                                              | 0.5346; 0.5372              | (0.3943, 0.7325); (0.3910, 0.7432)   |

Table 3: Estimated jump component  $K$ , and the corresponding mean and variance of jump size  $\mu_z, \sigma_z$ . We report the posterior mean/mode (for  $K$ ) and 90% posterior interval or range for  $K$ .

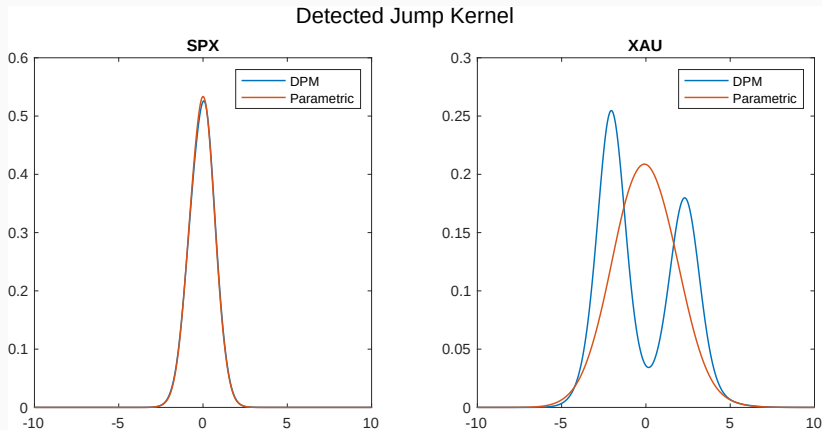


# Results I: Estimated jump intensity

| Estimated Jump Intensity for Financial Indices |                            |                            |                            |                            |
|------------------------------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
|                                                | SPX                        | XOI                        | TYX                        | XAU                        |
| Parametric                                     | 0.0918<br>(0.0673, 0.1225) | 0.1313<br>(0.0969, 0.1710) | 0.1398<br>(0.1040, 0.1799) | 0.0529<br>(0.0392, 0.0680) |
| DPM                                            | 0.0894<br>(0.0622, 0.1182) | 0.0961<br>(0.0666, 0.1291) | 0.1334<br>(0.0946, 0.1739) | 0.1191<br>(0.0933, 0.1484) |
| BNS test (1%)                                  | 0.0741                     | 0.0676                     | 0.1427                     | 0.0438                     |

Table 4: Post-convergence **posterior mean of jump intensity**  $\delta$  and their 90% posterior interval. The bottom row shows the Barndorff-Nielsen and Shephard (BNS) jump hypothesis test at 1% significance level. The bolded numbers are the estimated jump intensity that is closer to the BNS jump test results.

# Empirical Analysis: estimated jump kernel



**Figure 4:** The **estimated jump kernel** of both parametric and DPM jump models for four financial indices. The blue solid line represents the jump kernel estimated from DPM jump model. The red dashed line represents the estimated jump kernel from parametric jump model.

# Conclusion

The model we implemented here considers relatively simple structure of jumps:

- Jumps are i.i.d
- The jump size is not proportional to the size of volatility
- Do not consider volatility jumps either.

## **Further Investigation:**

- Comparing predictive density of parametric models and DPM jump models,
- Exploring further on the dynamics of extreme events to address the model misspecification mentioned above,
- Implication of the implied volatility smile when we considering multi-modes jumps.

**Questions?**

## References

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# References

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# Appendix

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## Implied Volatility: Definition

Implied volatility is the volatility that is implied by the market option price. Let  $c_t^m$  be market call option price and solve for  $\sigma = \hat{\sigma}$  such that

$$c_t^m = c_t^{BS}(\sigma), \quad (7)$$

where  $c_t^{BS}$  is the call option price calculated using Black-Scholes model (Black and Scholes, 1973).

# Models

Assume the stock price follows the structure as below:

1. SV model
2. Parametric SVJ model
3. SV model with mixture jumps

Let  $X$  be strike price,  $r_f$  be risk-free rate and  $T_m$  be time-to-maturity. The call option price can be simulated via Monte Carlo simulation as:

$$C(S_T, X, T_m) = \frac{1}{(1 + r_f)^{T_m}} E^Q[\max(S_{T_m} - X, 0)]. \quad (8)$$

# Implied Volatility Comparison

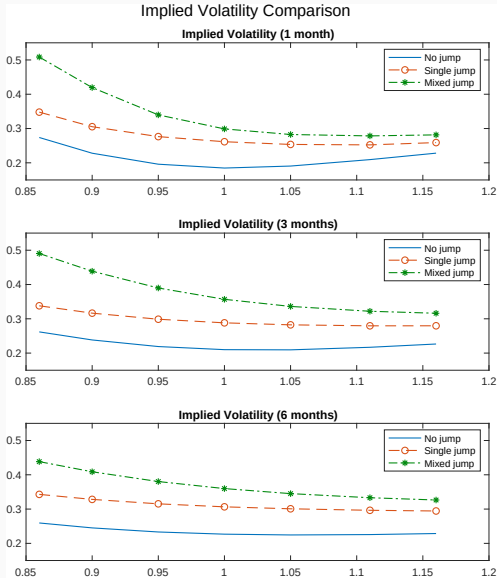


Figure 5: Implied volatility assuming the stock price follows the SV model, parametric SVJ model and SV-DPM jump model with moneyiness  $X/S_0$  ranging from 0.86 to 1.16 and time-to-maturity  $T_m$  being 1,3,6 months.

# DAGs with Slice Sampler

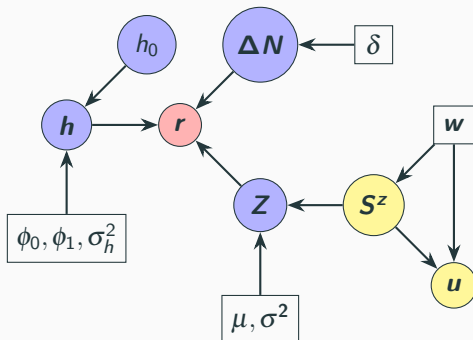


Figure 6: DAGs of the DPM jump model structure. Red circles represents observed data, purple circles are latent variables. Parameters are represented as rectangular. Yellow circles represents the auxiliary variables where  $u$  is introduced for slice sampler and  $S^z$  is an indicator functions for components.